$u_h^{n+1} = E_\Delta(\tau)^{n+1} u_h^0$. The matrix A defined in (3.10) is a proper representation of the operator E_Δ for a given (finite) grid size. It is easy to see that E_Δ is a linear operator.

We will generally assume that h and τ are not independent, but that instead there is a relation, say $h = h(\tau)$, such that $h \to 0$ if $\tau \to 0$. This justifies the notation $E_\Delta(\tau)$, i.e. characterizing the numerical scheme by a one-parameter family of operators.

3.4 Consistency

We now proceed with the analysis of difference schemes of the kind we have introduced in the previous section. Of course, the natural question is whether we are able to recover the true solution of our PDE problem at least asymptotically, using finer and finer meshes. For this analysis, we need to introduce a few new concepts. The first such concept is *concistency*. In the next section, we discuss *stability* and *convergence*. These are the basic paradigms used to analyze numerical schemes, and quantify the fidelity of our approximation.

Consistency basically means that at least asymptotically, the exact solution to our PDE problem satisfies the discrete equations. Let us first consider Model Problem 1 and Scheme 1. We define the consistency error[3] as the error obtained by substituting the exact solution of the PDE problem into the difference approximation. So let u be a smooth function that satisfies the heat equation. By the Taylor series analysis for the difference approximations (3.2) and (3.3), we have

$$\frac{u(x,t+\tau)-u(x,t)}{\tau} - a\frac{u(x-h,t)-2u(x,t)+u(x+h,t)}{h^2} = \underbrace{u_t(x,t) - au_{xx}(x,t)}_{=0} + \mathcal{O}(\tau) + \mathcal{O}(h^2) \tag{3.15}$$

The consistency error tends to zero, as both τ and h tend to zero. We say that the scheme is consistent with the heat equation. Re-write eq. (3.15) as

$$\frac{1}{\tau}\left\{u(x,t+\tau) - (\lambda u(x-h,t) + (1-2\lambda)u(x,t) + \lambda u(x+h,t))\right\} = \mathcal{O}(\tau) + \mathcal{O}(h^2) \tag{3.16}$$

where, as before, $\lambda = a\tau/h^2$. Note that the expression in curly brackets, when evaluated at the grid nodes, is just the error obtained after one step of the difference scheme. Also note that in our schemes for Model Problem 1, we enforce initial conditions and boundary conditions by direct injection (i.e. sampling the values at the grid nodes from the exact data), so these schemes are also consistent at the boundaries of the computational domain.

Let us now consider a general initial-boundary value problem. Consider, for some linear (spatial) differential operator L, the problem

$$u_t = Lu, \qquad x \in (0,1)\times(0,T] \tag{3.17}$$

$$u(\cdot,0) = u_0, \qquad x \in [0,1] \tag{3.18}$$

$$u(0,t) = u(1,t) = 0, \qquad t \in [0,T] \tag{3.19}$$

We assume that the problem is well-posed in some Banach space V.

For the general definitions to follow, we assume that a numerical scheme is defined by a family of linear evolution operators $E_\Delta(\tau) : V \to V$. Again, we shall always assume that there exists a relation between the mesh parameters τ and h, such that the mesh size can be characterized uniquely by τ (or h if that's more convenient). For fixed τ, such an operator maps the initial conditions via $(E_\Delta(\tau))^n u_0$ to a grid function at time instance $t^n = n\tau$. We give a formal definition of the consistency error as follows:

[3] In the literature, you also find the term *truncation error* instead of consistency error

Definition 3.1 *We say that a scheme for problem* (3.17)–(3.19)*, defined by the family of linear operators* $E_\Delta(\tau)$*, is consistent, if there exists a dense linear subspace* $V_c \subset V$*, such that for all* $u_0 \in V_c$ *and corresponding solution* u *one has*

$$\lim_{\tau\to 0}\left\|\frac{1}{\tau}\left(u(\cdot,t+\tau)-E_\Delta(\tau)u(\cdot,t)\right)\right\|_V = 0. \tag{3.20}$$

uniformly in $[0,T]$*. If we have for some* $p,q>0$*, and for all* t

$$\left\|\frac{1}{\tau}\left(u(\cdot,t+\tau)-E_\Delta(\tau)u(\cdot,t)\right)\right\|_V = \mathcal{O}(\tau^q+h^p), \tag{3.21}$$

we say the scheme is consistent of order p *in space and* q *in time.*

Note that we have again given the formal definition for a dense subspace $V_c \subset V$. For example, the consistency statements established by the Taylor series (cf. (3.2) and (3.3)) require a certain degree of smoothness, which may require us to choose functions from a class that does not coincide with V. Of course, the idea is to use density arguments, similar to that used in section 2.3 to extend results to all initial data $u_0 \in V$ in the end. We come back to this below.

We have assumed that a relation between τ and h exists. Often this is of the form $\tau = \mathcal{O}(h^r)$. In this case, we may express the order of consistency as a function of the spatial grid size only, and simply say that the scheme is consistent of order $h^{\min(p,rq)}$. Such relations between τ and h are often chosen to maximize the accuracy, or they are required for stability, which we discuss next.

3.5 Stability and Convergence

The basic premise of our exposition of stability and convergence is that we carry our a sequence of calculations on successively finer meshes. This means that we suppose (an ultimately infinite) sequence of grid parameters $\{\tau_k\}$ and $\{h_k\}$, which both tend to zero as $k \to \infty$. The general idea is that we should recover the exact solution of our PDE problem in the limit. It may be tempting to think that a consistent scheme necessarily produces the correct solution in this limit. We will see that this is not automatically guaranteed: Consistency merely tells us that one step of the scheme (or, equivalently, finitely many steps) are ok. But in the limit of mesh refinement, we need to carry out infinitely many steps to reach any finite time $t > 0$. In that limit process, consistency alone is not enough to guarantee convergence, and we shall establish *stability* as the missing link.

To illustrate the concept we carry out a case study using Model Problem 1 and the explicit Scheme 1, before we deal with the abstract theory. First, we define the error of the scheme as the difference between the exact solution and the numerical approximation. Again we have the choice of interpreting the error as being defined only at the grid nodes, or as a grid function. For the purposes of this first example we choose the former interpretation. So define

$$e_i^n := u_i^n - u(x_i,t^n), \tag{3.22}$$

where $u(x,t)$ is the exact solution to Model Problem 1. Using direct injection of the correct boundary data, we can set $e_0^n = e_N^n = 0$. Then we can re-write the error (3.22) for $i = 1,\dots N-1$ as

$$\begin{aligned} e_i^n &= u_i^n - u(x_i,t^n) \\ &= \lambda e_{i-1}^{n-1} + (1-2\lambda)e_i^{n-1} + \lambda e_{i+1}^{n-1} - \tau r_i^{n-1}, \end{aligned}$$

where we have used (3.5) from the definition of Scheme 1, as well as the consistency error (3.16). The quantity r_i^{n-1} is defined as the left-hand side of (3.16), evaluated at $(x,t) = (x_i,t^{n-1})$.

Again, we may set $r_0^n = r_N^n = 0$. Let us assume that $0 \le \lambda \le \frac{1}{2}$. Then

$$\begin{aligned} |e_i^n| &\le \lambda|e_{i-1}^{n-1}| + (1-2\lambda)|e_i^{n-1}| + \lambda|e_{i+1}^{n-1}| + \tau|r_i^{n-1}| \\ &\le (\lambda + 1 - 2\lambda + \lambda) \max_{0\le i\le N} |e_i^{n-1}| + \tau \max_{0\le i\le N} |r_i^{n-1}| \end{aligned}$$

Note that the inequality holds for all i, since the right-hand side does not depend on i anymore. In particular, we can take the maximum over i on the left-hand side, and we have by recursion

$$\max_{0\le i\le N} |e_i^n| \quad \le \quad \max_{0\le i\le N} |e_i^0| + \tau \sum_{l=0}^{n-1} \max_{0\le i\le N} |r_i^l|$$

If we sample the initial conditions from the exact solution, as in Scheme 1, we have $e_i^0 = 0$. Furthermore, we have

$$\begin{aligned} \max_{0\le i\le N} |e_i^n| &\le \tau n \max_{\substack{0\le i\le N \\ 0\le l\le n-1}} |r_i^l| \\ &\le TC(\tau + h^2) \end{aligned} \tag{3.23}$$

where in the last inequality, we have used $n\tau \le T$, and the expression for the consistency error (3.16). Since

$$\lambda = \frac{a\tau}{h^2} \le \frac{1}{2} \qquad \Rightarrow \tau = \mathcal{O}(h^2),$$

we have, noting that the estimate (3.23) holds for all n, and all $n\tau \le T$,

$$\max_{\substack{0\le i\le N \\ n\tau\le T}} |e_h^n|_\infty \le CTh^2.$$

The error tends to zero with same order as the consistency error, provided $\lambda \le 1/2$. We say that the scheme *converges*. It can be shown by counterexample that $\lambda > 1/2$ can lead to blow-up of the solution for certain initial conditions. Consequently, the condition $\lambda \le 1/2$ is not only sufficient but also necessary. Such conditions often play a major role in the analysis of numerical schemes. They ensure the *stability* of a scheme.

We now give a formal definition of stability and convergence.

Definition 3.2 *A difference scheme for problem* (3.17)–(3.19)*, defined by a family of operators $E_\Delta(\tau) : V \to V$ is said to be stable with respect to the norm $\|\cdot\|_V$, if for any $T > 0$ there is a constant $C = C(T)$, depending only on T, such that*

$$\|E_\Delta(\tau)^n\|_V \le C(T), \tag{3.24}$$

for all n, τ, such that $0 < \tau < \tau_0$ and $n\tau \le T$.

Please note that the superscript n in (3.24) is a *power*, not a time index! Note that the definition of stability automatically implies robustness with respect to the initial conditions, because the operator norm is defined as

$$\|E_\Delta(\tau)^n\| = \sup_{\substack{u\in V \\ u\ne 0}} \frac{\|E_\Delta(\tau)^n u\|}{\|u\|}, \tag{3.25}$$

which guarantees that the norm of the numerical solution $E_\Delta(\tau)^n u_0$ remains bounded for any $u_0 \in V$. Note, however, that the definition of stability is always with respect to the norm $\|\cdot\|_V$. We have so far used only the maximum norm for our model problem, but in general the choice of norm is not unique, and a given scheme may very well be stable with respect to one norm, but not with respect to another.

Now we define convergence:

Definition 3.3 *Let $T > 0$ be fixed. We say a difference scheme for problem* (3.17)–(3.19)*, defined by a family of operators $E_\Delta(\tau) : V \to V$ is* convergent *with respect to the norm $\|\cdot\|_V$, provided that for any $u_0 \in V$, and for all $t \in [0, T]$,*

$$\lim_{j\to\infty} \left\| E_{\tau_j}^{m_j} u_0 - u(\cdot, t) \right\|_V \to 0, \tag{3.26}$$

where $\{m_j\}$ and $\{\tau_j\}$ are sequences such that $\tau_j \to 0$ and $m_j\tau_j \to t$. The order of convergence *is defined in a manner analogously to* (3.21)*.*

3.6 Lax Equivalence Theorem

In the previous section we saw that a stability condition was crucial for establishing convergence of the explicit scheme for our Model Problem 1. Stability is always needed for convergence, as the celebrated Lax-Equivalence Theorem states.

Theorem 3.1 (Lax-Equivalence Theorem) *Assume that problem* (3.17)–(3.19) *is well posed. Let $E_\Delta(\tau) : V \to V$ be a consistent linear scheme. The scheme is convergent if and only if it is stable. Furthermore, for $u_0 \in V_c$, with V_c as in Definition 3.1, stability implies that the order of consistency is equal to the order of convergence.*

Proof In the following, we prove that stability implies convergence. (The reverse implication is easy: If a scheme is convergent, it must be stable, for otherwise we could choose initial data, and a mesh refinement sequence, for which the discrete solution blows up in finite time, which would contradict the assumption of convergence.)

To save some writing define $u(\cdot, t) =: \mathbf{u}(t) \in V$. Assume that n and τ are such that $n\tau \le T$, and consider, for some $t \in [0, T]$

$$\begin{aligned} E_\Delta(\tau)^n u_0 - \mathbf{u}(t) = E_\Delta(\tau)^n u_0 &\mp E_\Delta(\tau)^{n-1}\mathbf{u}(\tau) \\ &\mp E_\Delta(\tau)^{n-2}\mathbf{u}(2\tau) \\ &\vdots \\ &\mp E_\Delta(\tau)^{n-k}\mathbf{u}(k\tau) \\ &\vdots \\ &\mp \mathbf{u}(n\tau) - \mathbf{u}(t) \end{aligned}$$

Note that we have introduced a number of extra terms that cancel each other. The reason for adding these terms is that we can now write the error in terms of the consistency error at each step. Indeed, we may re-arrange the sum to produce

$$E_\Delta(\tau)^n u_0 - \mathbf{u}(t) = \sum_{k=0}^{n-1} E_\Delta(\tau)^k \left\{ E_\Delta \mathbf{u}((n-k-1)\tau) - \mathbf{u}((n-k)\tau)) \right\} + \mathbf{u}(n\tau) - \mathbf{u}(t).$$

Note that the term in the curly brackets is (up to a factor of τ) the consistency error.

Taking the norm, and using the triangle inequality, we have

$$
\begin{aligned}
&\|E_\Delta(\tau)^n u_0 - \mathbf{u}(t)\| \\
&\le \sum_{k=0}^{n-1} \left\|E_\Delta(\tau)^k\right\| \|E_\Delta \mathbf{u}((n-k-1)\tau) - \mathbf{u}((n-k)\tau))\| + \|\mathbf{u}(n\tau) - \mathbf{u}(t)\| \\
&\le \max_{0\le k\le n-1} \left\|E_\Delta(\tau)^k\right\| \tau \sum_{k=0}^{n-1} \left\| \frac{1}{\tau} \left(E_\Delta \mathbf{u}((n-k-1)\tau) - \mathbf{u}((n-k)\tau))\right)\right\| + \|\mathbf{u}(n\tau) - \mathbf{u}(t)\| \\
&\le \max_{0\le k\le n-1} \left\|E_\Delta(\tau)^k\right\| n\tau \max_{0\le k\le n-1} \left\| \frac{1}{\tau} \left(E_\Delta \mathbf{u}((n-k-1)\tau) - \mathbf{u}((n-k)\tau))\right)\right\| + \|\mathbf{u}(n\tau) - \mathbf{u}(t)\| \\
&\le \underbrace{\sup_{\substack{0<\tau<\tau_0 \\ n\tau\le T}} \|E_\Delta(\tau)^n\|}_{\le C(T)} T \underbrace{\sup_{\substack{t\in(0,T) \\ t+\tau\le T}} \left\| \frac{1}{\tau} \left(E_\Delta \mathbf{u}(t) - \mathbf{u}(t+\tau))\right)\right\|}_{\to 0 \text{ (consistency)}} + \underbrace{\|\mathbf{u}(n\tau) - \mathbf{u}(t)\|}_{\to 0 \text{ as } n\tau\to t}
\end{aligned}
$$

Now, because the operator norm is uniformly bounded, and the scheme is consistent, we see immediately that under the required limit $\tau \to 0$, and $n\tau \to t$, the error converges to zero, at least for those solutions in $V_c \subset V$, for which the consistency estimate holds. We can extend the convergence to all solutions in V by a density argument: Let $\{u_{0,k}\} \subset V_c$ be a sequence that converges to $u_0 \in V$, and let $\mathbf{u}(t) = E(t)u_0$ be the exact solution corresponding to the limit. Then we have

$$E_\Delta(\tau)^n u_0 - \mathbf{u}(t) = E_\Delta(\tau)^n u_0 \pm E_\Delta(\tau)^n u_{0,k} \pm E(t)u_{0,k} - E(t)u_0$$

By triangle inequality, we have

$$\|E_\Delta(\tau)^n u_0 - \mathbf{u}(t)\| = \|E_\Delta(\tau)^n (u_0 - u_{0,k})\| + \|E_\Delta(\tau)^n u_{0,k} - E(t)u_{0,k}\| + \|E(t)(u_{0,k} - u_0)\|$$

By stability of the numerical scheme, and the well-posedness of the PDE problem, we can bound the first term and the last term as

$$\|E_\Delta(\tau)^n (u_0 - u_{0,k})\| + \|E(t)(u_{0,k} - u_0)\| \le c\,\|u_{0,k} - u_0\|$$

We can make these terms as small as we please, since $u_{0,k} \to u_0$, say smaller than arbitrary $\varepsilon > 0$, by choosing k large enough. For such a k, we can bound the second term in (3.27) by the convergence result we have already derived for $u_{0,k} \in V_c$, i.e.

$$\|E_\Delta(\tau)^n u_{0,k} - E(t)u_{0,k}\| < \varepsilon$$

by choosing $|n\tau - t|$ small enough. (Recall that we take the limit $\tau \to 0$ and $n\tau \to t$.) This establishes convergence for the general case $u_0 \in V$.

It is an easy corollary that the order of convergence is the same as the consistency error for smooth enough solutions. ■

3.7 Further Reading

The general approach follows the classic text by Richtmyer and Morton [11], Chapter 3, and the later text by Atkinson and Han [1], Chapter 5. In the former reference, the extension to inhomogeneous problems is discussed.